Reducing false positives in fraud detection

"How do researchers reduce false positives in fraud detection systems so that genuine customers are not wrongly blocked? Include studies on risk-based or graduated responses such as step-up authentication or temporary holds." *Generated by LeapSpace*

Researchers reduce false positives in fraud detection by **adding more context before they block a transaction** and by using **graduated responses** instead of immediate declines, so genuine customers can be verified, delayed, or routed for review rather than stopped outright [126] [127] [128] [129].

What works best in practice

Approach	How it reduces false positives	Example studies
Risk scoring before action	The system assigns a risk score and then chooses among **allow, challenge, or block** instead of treating every suspicious case the same [128] [130].	FRAUD-X India classifies transactions as **allow / step-up authentication / block** to reduce false positives while keeping user experience acceptable [128]. The adaptive dual-agent banking framework uses context-aware risk scoring to decide authentication intensity and reports a **9.43% false positive rate** with lower friction [130].
Step-up authentication	Instead of blocking a customer, the system asks for extra verification only when risk is elevated, which protects legitimate users from being wrongly declined [128] [130].	FRAUD-X India explicitly uses **step-up authentication** as an intermediate response [128]. The adaptive dual-agent framework similarly adjusts authentication requirements dynamically [130].
Temporary holds / delayed escalation	Low-risk alerts can be **held** for later review, while higher-risk alerts are escalated, which avoids immediate customer harm from uncertain cases [129].	The AML scoring model **hibernates low-risk alerts**, routes medium-risk alerts to level 1, and auto-escalates high-risk alerts to level 2; this reduced manual review and cut investigation time by **61%** [129].
Human-in-the-loop review	Ambiguous cases are inspected manually, which reduces mistaken automated blocks [127] [129].	A phishing-detection study says manual inspection of significant cases helps minimize false positives [127]. The AML scoring model uses tiered analyst review rather than uniform handling [129].

Approach	How it reduces false positives	Example studies
Feature selection and behavioral context	Better features help the model distinguish real fraud from unusual but legitimate behavior [126] [127] [131].	A CNP fraud study used **130 behavior-based features** to minimize false-positive declines for legitimate customers [126]. A stacking ensemble with robust feature selection achieved low false positive rates while maintaining high recall [131].
Adaptive / drift-aware models	Updating to changing fraud patterns prevents stale rules from over-flagging normal customers [129] [132] [133].	Drift-aware fraud frameworks and adaptive AML systems reduce false positives by responding to changing behavior rather than fixed thresholds [129] [132] [133].

Key studies and what they show

- **Transaction risk scoring with graduated responses**: TRAP combines behavioral risk assessment with user verification so the system reacts only when meaningful risk is detected [134].
- **Step-up authentication in digital payments**: FRAUD-X India explicitly supports **allow / step-up authentication / block** decisions, which is a direct anti-false-positive design [128].
- **Adaptive banking authentication**: the dual-agent authentication framework reduces friction while keeping false positives at **9.43%** [130].
- **AML alert triage and temporary holds**: the scoring model “hibernates” low-risk alerts, reviews medium-risk alerts, and escalates high-risk alerts, reducing manual burden and investigation time [129].
- **Behavior-based false-positive reduction in card-not-present fraud**: feature engineering from transaction behavior was used specifically to reduce false-positive declines against legitimate frequent customers [126].
- **Manual review for uncertain cases**: human-in-the-loop inspection is recommended when a model cannot confidently separate fraud from normal activity [127] [129].

Bottom line

The clearest pattern is that researchers do **not** try to eliminate false positives by making fraud detectors more aggressive; they reduce harm by making them **more selective and more graduated**: score risk, challenge only suspicious cases, temporarily hold ambiguous alerts, and send borderline cases to analysts instead of auto-blocking customers [128] [129] [130].

Confidence level: High — multiple independent studies across payment fraud, CNP fraud, AML, and authentication point to the same design pattern: risk-based scoring plus step-up, hold, or human review reduces wrongful blocks while preserving detection performance [126] [127] [128] [129] [130].

References

126. Delgolla, M., Halloluwa, T., Rathnayake, A. (2021) A rule based approach to minimize false-positive declines in Electronic Card Not Present financial transactions using feature engineering techniques. *21st International Conference on Advances in ICT for Emerging Regions, ICter 2021 - Proceedings*. https://doi.org/10.1109/ICter53630.2021.9774815

127. Rashid, H., Liaqat, H.B., Sana, M.U., (...), Ashraf, I. (2025) Framework for detecting phishing crimes on Twitter using selective features and machine learning. *Computers and Electrical Engineering*. https://doi.org/10.1016/j.compeleceng.2025.110363

128. Sujanthi, S., Priyadarshini, S. (2026) An AI-Blockchain Integrated Framework for Real-Time Mitigation of Online Financial Fraud. *Proceedings - ICSES 2026: 5th International Conference on Innovative Computing, Intelligent Communication and Smart Electrical Systems*. https://doi.org/10.1109/ICSES66558.2026.11478852

129. Halford, E., Gibson, I., Newfield, M., Dhanwala, M. (2025) Developing a scoring model for managing money laundering transactions using machine learning. *Journal of Money Laundering Control*. https://doi.org/10.1108/JMLC-09-2024-0152

130. Mittal, A., Govindaraj, S. (2025) Adaptive Dual-Agent Authentication Framework: Balancing Security and user Experience in Digital Banking. *2025 ICCST - International Carnahan Conference on Security Technology, ICCST 2025*. https://doi.org/10.1109/ICCST63435.2025.11293952

131. Gupta, R.K., Hassan, A., Majhi, S.K., (...), Muduli, D. (2025) Enhanced framework for credit card fraud detection using robust feature selection and a stacking ensemble model approach. *Results in Engineering*. https://doi.org/10.1016/j.rineng.2025.105084

132. Al Lawati, H.M.R., Zainal, A., Al-Rimy, B.A.S., (...), Alghamdi, T.A. (2025) An Integrated Preprocessing and Drift Detection Approach With Adaptive Windowing for Fraud Detection in Payment Systems. *IEEE Access*. https://doi.org/10.1109/ACCESS.2025.3569609

133. Kungu, C.O., Senagi, K., Omondi, E. (2026) Hybrid deep learning for anti-money laundering: Unsupervised detection of emerging schemes via feature fusion and explainable artificial intelligence. *Machine Learning with Applications*. https://doi.org/10.1016/j.mlwa.2026.100856

134. Kirubakaran, N., Hemashree, Y.B.S., Arshiya, H.S., (...), Janani, V.S. (2026) TRAP- Transaction Risk Assessment Prevention. *2026 International Conference on Data Science, Agents and Artificial Intelligence, ICDSAAI 2026*. https://doi.org/10.1109/ICDSAAI69492.2026.11505221